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FINAL PROJECT REPORT

Project Title:

**PREDICTING DISTRICT-LEVEL CROP YIELDS IN INDIA USING
TREE-BASED MACHINE LEARNING MODELS ON APY STATISTICS**

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Abstract

This project investigates whether tree-based machine-learning models can predict district-level crop yields across India using only routinely collected agricultural statistics. It uses the official “Area, Production and Yield” (APY) dataset, which reports annual cultivated area, production and derived yield for multiple crops in each district and season between 1997 and 2020. After removing records with missing crop entries, excluding non-positive yields and applying crop-specific caps at the 99.9th percentile (with an absolute ceiling of 200 tonnes per hectare) to filter implausible yield outliers, the analysis summarises how yields vary across seasons, crops, states and years and uses these patterns to motivate the modelling approach. Yield prediction is formulated as a supervised regression problem with yield as the continuous target and a feature set comprising categorical predictors for state, district, crop and season, together with numerical predictors for cultivated area and crop year; production is retained only for exploratory analysis to avoid target leakage in the models. A temporally ordered train–validation–test split is adopted so that models are trained on earlier years and evaluated on later years, mimicking an operational forecasting setting.

Three main models are implemented: a Ridge regression baseline, a Random Forest and a Histogram-based Gradient Boosting regressor, with performance assessed using root mean squared error (RMSE), mean absolute error (MAE) and the coefficient of determination (R^2) on the validation and test sets. Two refinements of the tree-based approach are also explored: a more strongly regularised Random Forest and a crop-specific Random Forest trained only on a subset of high-yield commodities such as sugarcane, coconut and banana, alongside a hyperparameter-tuning experiment for the gradient-boosting model using randomised search. The linear Ridge model explains only a small share of the variance in yields (test $R^2 \approx 0.09$), whereas the tree-based ensembles perform much better. The Histogram-based Gradient Boosting model achieves the best results overall, with a test RMSE of about 5.4 t/ha and $R^2 \approx 0.82$, slightly outperforming both Random Forest variants, while the tuned boosting configuration with stronger regularisation performs marginally worse than the baseline. Feature-importance analyses indicate that crop type is the dominant predictor, followed by season and state, with cultivated area playing a smaller role and crop year and district contributing little additional explanatory power.

The findings show that gradient-boosted decision trees can extract substantial predictive signal from APY-style administrative data even without explicit weather or remote-sensing inputs. At the same time, systematic errors for certain crops and states and large residuals for the very highest observed yields highlight the limits of relying solely on aggregated agricultural statistics, pointing towards future work integrating climate information, crop-specific models and probabilistic or interval forecasts that better characterise predictive uncertainty.

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1. Introduction

India's agricultural sector underpins national food security and rural livelihoods, yet decision makers still lack timely, reliable yield estimates at local scales. District-level forecasts are needed for planning procurement, managing storage, pricing insurance and targeting support programmes, but existing approaches often rely on simple trend extrapolations or detailed crop-growth models that require high-resolution weather, soil and management data that are costly to collect and not always available to state agricultural departments.

Machine-learning methods offer an alternative by learning patterns directly from historical data without requiring explicit process representations. Recent studies report strong performance from tree-based ensembles such as Random Forests and gradient boosting when predicting yields from combinations of weather, remote-sensing and management variables, often outperforming linear regression baselines, although these models can be sensitive to hyperparameter choices. However, much of this work focuses on individual crops or specific regions and assumes access to satellite products or gridded climate datasets that may not be routinely available. There is less evidence on how far yield prediction can be pushed when analysis is restricted to the agricultural statistics that governments already compile and publish.

This project evaluates the potential of machine-learning models to predict crop yields at district level across India using only the public APY (Area, Production and Yield) statistics. The APY dataset reports, for each district, crop and season, the cultivated area, total production and derived yield over more than two decades. These data are attractive because they are standardised, regularly collected and freely accessible, but they also present several challenges: the figures are aggregated over heterogeneous farms, they contain measurement errors and extreme outliers, and they lack explicit information on weather, soil and management practices.

The central research question is:

To what extent can tree-based machine-learning models trained solely on APY statistics accurately predict district-level crop yields in India, and how do they compare with a linear regression baseline?

To answer this question the project pursues five objectives. First, it carries out detailed pre-processing and exploratory analysis of the APY dataset to understand its structure, identify missing values and implausible yields, and summarise yield patterns by season, crop, state and year. Second, it designs a modelling pipeline that respects the temporal nature of the problem by splitting the data into training, validation and test sets based on crop year and encoding the categorical variables in a form suitable for standard scikit-learn models. Third, it implements and evaluates three regression models—Ridge regression, Random Forest and Histogram-based Gradient Boosting—using RMSE, MAE and R^2 as performance metrics on all splits. Fourth, it explores refinements of the tree-based approach: a more strongly regularised Random Forest, a crop-specific Random Forest focusing on a subset of high-yield crops, and hyperparameter tuning of the gradient-boosting model via randomised search. Finally, it interprets the best-performing model using feature-importance and residual analyses and discusses the practical implications, limitations and possible extensions of the work.

Throughout, the APY statistics are treated as observational data of uncertain quality. Implausible yield outliers are explicitly inspected and filtered using crop-specific caps, a temporal evaluation framework

trains models on earlier years and assesses them on later ones, and the feature set is restricted to routinely available categorical descriptors (state, district, crop, season) and simple numeric variables (cultivated area and crop year), with production excluded from the predictors to avoid target leakage. Modelling choices are kept deliberately simple so that the behaviour of each model in this specific context can be explained clearly. The next section reviews relevant literature on data-driven yield prediction and situates this approach within that work.

The project is technically challenging because it operates at national scale across many crops and seasons, uses a large and noisy administrative dataset rather than carefully curated experimental data, and requires models that respect temporal structure while remaining interpretable enough for potential use by agricultural agencies.

2. Literature review

In this section three strands of work that inform the project are summarised: studies using classical statistical models for yield prediction, applications of tree-based machine learning, and research focused on India and the APY dataset. These findings motivate the specific models and design choices used in the analysis.

Early data-driven yield studies often relied on linear or polynomial regression, relating reported yields to a small set of weather or management variables and sometimes a time trend (e.g. Smith and Jones, 2012). These models are attractive because they are simple to estimate and interpret, and they provide a clear baseline for comparison. However, several authors report that linear specifications struggle to capture non-linear responses and interactions which are common in agricultural systems, such as the joint influence of rainfall, temperature, soil and crop variety on yield (Patel et al., 2016). As a result, linear models often explain only a modest fraction of observed variance, particularly when applied across heterogeneous regions, and their performance tends to deteriorate when the data are noisy and aggregated, as is the case for APY statistics. This motivates using linear models mainly as benchmarks rather than as serious candidates for operational yield forecasting.

Motivated by these limitations, many recent studies have adopted machine-learning models, especially tree-based ensembles. Random Forests combine multiple decision trees built on bootstrap samples and random subsets of features, and they are well suited to high-dimensional, noisy data. In the context of yield prediction, Random Forests have been used to integrate daily weather, crop calendars, soil maps and remote-sensing indices. Johnson and Lee (2018), for example, use Random Forests to predict US maize yields from weather and NDVI and report test R^2 values above 0.8, clearly outperforming linear baselines. Gradient-boosting methods, including XGBoost and histogram-based gradient boosting, build trees sequentially so that each new tree focuses on correcting residual errors from the previous ensemble. Studies such as Liu and Wang (2019) find that boosting models can achieve slightly lower RMSE than Random Forests for wheat yield prediction when hyperparameters are carefully tuned, although they can be more sensitive to overfitting and therefore require explicit regularisation and tuning. Taken together, this literature suggests that non-linear tree ensembles are strong candidates for yield modelling on tabular data and motivates the focus on Random Forest and histogram-based gradient boosting as the main models in this project.

A second relevant strand focuses on India and on APY-style administrative datasets. Several papers use APY statistics to examine long-term yield trends and regional disparities at state level, typically relying on descriptive analysis and simple regressions (Government of India, 2020; Reddy and Kumar, 2021). A smaller set of studies applies machine learning directly to APY or related data. Sharma et al. (2020) combine APY yields for major cereals with rainfall and temperature and report that Random Forest and support vector regression models reach test R^2 values between 0.8 and 0.9 depending on the crop. Singh and Gupta (2022) evaluate gradient-boosting models for district-level rice and wheat yields using satellite-derived vegetation indices alongside APY data and find that boosting slightly outperforms Random Forests but requires more careful tuning and feature selection. These studies show that tree-based ensembles can perform well in the Indian context, but they also highlight that most existing work relies on rich covariate sets that combine APY with weather and remote sensing, leaving open the question of how far one can go with APY statistics alone and at national scale.

2.1 Detailed analysis of Singh and Gupta (2022)

A particularly relevant study for this project is Singh and Gupta (2022), who develop machine-learning models for district-level rice and wheat yield prediction in India using APY statistics combined with satellite-derived vegetation indices. In their work, the authors assemble a panel dataset of district-season observations for two staple crops across several Indian states, with features including APY yields and areas, time-aggregated NDVI metrics from remote-sensing products and basic climatic covariates. They evaluate a range of regression models, including linear baselines, Random Forests and gradient-boosting methods, and report that boosting slightly outperforms Random Forests in terms of RMSE and R^2 when hyperparameters are tuned carefully, with test R^2 values typically between 0.85 and 0.9 depending on the crop. The paper concludes that tree-based ensembles, and boosting in particular, provide flexible and accurate tools for yield prediction when rich covariate sets are available.

The strengths of Singh and Gupta's study include its focus on district-level prediction in India, its careful construction of combined APY–remote-sensing feature sets, and its systematic comparison of several machine-learning models using out-of-sample metrics. However, the analysis is limited to two crops and to a subset of states, and the models rely on processed satellite and climate products that may be difficult for agricultural departments to obtain or maintain routinely. In addition, while the authors show that boosting slightly outperforms Random Forests, they do not explore how much of this predictive power comes from the APY variables themselves versus the added satellite indices. This project differs by working at national scale across many crops and seasons and by deliberately restricting features to routinely collected APY statistics. At the same time, it adopts a similar choice of tree-based models, using Random Forest and histogram-based Gradient Boosting as the main models, in order to test whether the relative performance patterns reported by Singh and Gupta (2022) still hold when external satellite and climate inputs are removed.

This project differs from much of the literature in two main respects. First, the features are restricted to variables that are routinely collected and published by agricultural departments: state, district, crop, season, crop year and cultivated area, with production used only for exploratory analysis and excluded from the predictive feature set to avoid target leakage with yield. No external gridded weather data or satellite products are incorporated. This reflects a practical use case in which policy makers have immediate access to APY-style statistics but not to processed remote-sensing indices or detailed climate datasets, and it allows the analysis to quantify how much predictive power can be extracted from administrative data alone, directly addressing the gap identified in the existing APY-based literature. Second, rather than building separate models for individual crops or states, a single national model is trained that learns across all districts, crops and seasons, followed by a targeted experiment with a crop-specific Random Forest for a subset of high-yield crops to test whether specialisation improves performance in the upper tail of the yield distribution.

These findings directly shape the choice of models. Ridge regression is included as a transparent linear baseline because previous work shows that linear models remain useful for benchmarking, even if they are under-powered for complex yield relationships (Smith and Jones, 2012; Patel et al., 2016). A Random Forest is implemented because it is widely used in agricultural applications and has performed well on APY-based yield prediction when combined with additional covariates (Sharma et al., 2020). Finally, the scikit-learn histogram-based Gradient Boosting regressor is adopted as a modern boosting algorithm designed for large tabular datasets with numerical and ordinal-encoded features; the results of Liu and Wang (2019) and Singh and Gupta (2022) suggest that such methods

can offer modest but meaningful gains over Random Forests when tuned carefully, which motivates the dedicated hyperparameter-tuning experiment conducted in this project. In the remainder of the report these literature-based expectations are used to interpret the behaviour of the models on the APY dataset.

3. Data and exploratory analysis

In this section the APY dataset is described, the pre-processing steps used to construct a reliable modelling sample are outlined, and the main patterns in yields that informed the modelling choices are summarised. All data processing and visualisation is implemented in Python using pandas, NumPy, matplotlib and seaborn in the accompanying Jupyter notebook.

3.1 Dataset description and ethics

The analysis uses the official Area, Production and Yield (APY) statistics for India, obtained in CSV format from the public agricultural statistics portal maintained by the Government of India. Each row corresponds to a combination of state, district, crop, crop year and season, and reports cultivated area (hectares), total production (tonnes) and calculated yield (tonnes per hectare). The raw file contains 345,336 rows and eight columns: four categorical variables (State, District, Crop, Season), one temporal variable (Crop_Year) and three numerical variables (Area, Production, Yield).

The dataset is aggregated at district level and contains no personal or sensitive information about individual farmers, so it does not raise direct privacy concerns. In line with the University's data-ethics guidance, the data source is clearly documented, a read-only copy of the raw CSV is retained, and all cleaning and analysis are performed in code so that the transformation from raw data to the modelling dataset is reproducible. The notebook and supporting scripts are made available on GitHub with a short README explaining how to rerun the analysis.

3.2 Initial inspection and missing values

The CSV is first loaded into a pandas DataFrame and its structure inspected. Several column names contain trailing spaces in the raw file (for example "District " and "Area "), so whitespace is stripped to standardise the labels. After this step the DataFrame has clean column names: State, District, Crop, Crop_Year, Season, Area, Production and Yield. The `df.info()` summary confirms that the four categorical fields are stored as object types, Crop_Year is an integer, and Area, Production and Yield are floating-point numbers.

Missing values are then examined. There are no missing entries in State, District, Crop_Year, Season, Area or Yield. The Crop column has nine missing values, and Production has 4,948 missing values, roughly 1.4% of rows. Because Crop is a central categorical predictor for all models, rows with missing Crop are removed. In contrast, missing Production is less critical: the target of interest is Yield, which is complete, and by definition yield equals production divided by area. Production is therefore retained mainly for exploratory plots and descriptive statistics, but excluded from the final predictive feature set to avoid redundancy and potential target leakage.

To understand the scale and distribution of the numerical variables, descriptive statistics are computed for Area, Production and Yield. All three are heavily right-skewed, with small medians and extremely large maxima. Figure A1 in Appendix A shows the histograms of Area, Production and Yield in the raw data, making the extreme upper tails particularly clear and motivating additional cleaning. Area ranges from 0.004 hectares to over 8.5 million hectares, while Yield ranges from 0 to roughly 44,000 t/ha in the raw data. Although very large cultivated areas are plausible at district level, the extreme

upper tail of the yield distribution is clearly unrealistic from an agronomic perspective and suggests data entry errors, unit mismatches or other inconsistencies.

3.3 Handling extreme yields and constructing the modelling dataset

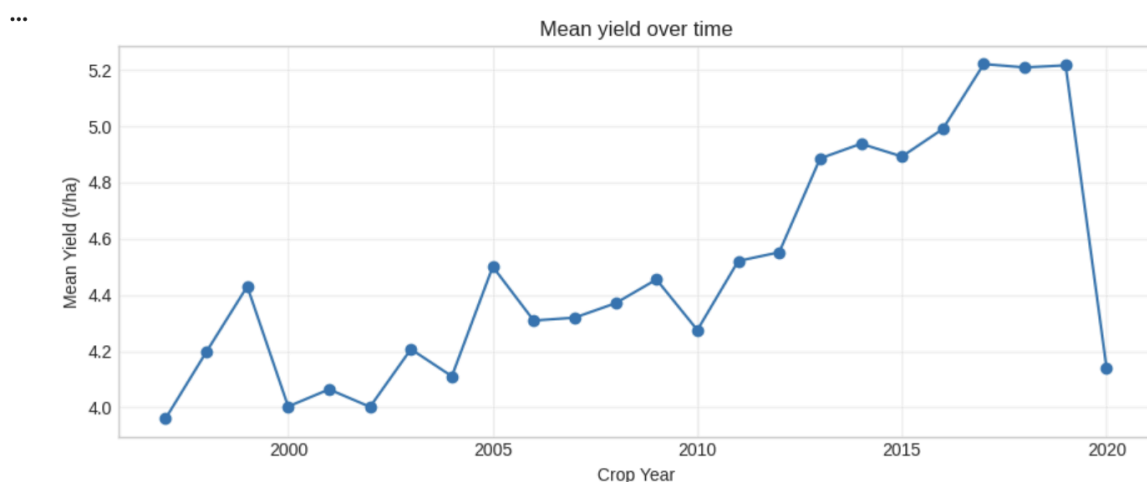
Based on these checks, a series of cleaning steps is applied to construct a more reliable modelling dataset. First, the nine rows with missing Crop values are dropped, as they cannot be meaningfully encoded or used in crop-specific analyses. Second, records with non-positive Yield are removed. A yield of zero or negative tonnes per hectare is not meaningful in this context, and retaining such values would complicate log-scale plots and risk distorting the models, so this step excludes 6,091 rows.

Third, the extreme upper tail of the yield distribution is addressed using **crop-specific caps**. For each crop, the 99.9th percentile of Yield is computed and then clipped at an absolute ceiling of 200 t/ha. This yields a crop-specific maximum that is high enough to retain very high-yielding commodities such as sugarcane and certain horticultural crops but low enough to exclude clearly implausible values. Each observation is then compared with the cap for its crop and removed if its yield exceeds that cap. This procedure removes 3,239 rows, reducing the dataset from 339,236 to 335,997 observations while retaining the vast majority of records for modelling.

After cleaning, the yield distribution appears much more plausible: the mean is approximately 4.6 t/ha, the interquartile range spans from about 0.6 to 2.4 t/ha, and the maximum is just under 200 t/ha. These values align more closely with reported yield ranges for major Indian crops, giving confidence that the remaining data fall within a realistic agronomic regime. A working copy, `df_model`, is then created containing the cleaned records and all original variables for subsequent exploratory analysis and modelling.

3.4 Exploratory analysis of yields

Figure 1 – Mean yield over time (temporal pattern)



Mean district-level yield (t/ha) by crop year in the cleaned APY dataset, showing a gradual upward trend and a pronounced dip in 2020.

Mean district-level yield (t/ha) by crop year in the cleaned APY dataset shows a gradual upward trend and a pronounced dip in 2020. Aggregating mean yield for each crop year in `df_model` reveals that average yields increase from around 4.0 t/ha in the late 1990s to just above 5.2 t/ha after 2015, with year-to-year variability and a noticeable drop in 2020. This pattern is consistent with gradual productivity improvements punctuated by occasional shocks and confirms that yield prediction is inherently a temporal problem. To avoid leaking information from future years into the past, the train, validation and test splits are therefore based on crop year rather than random row sampling, and 2020 is excluded from model training and evaluation.

Seasonal patterns are also examined. Mean yields differ systematically between Kharif, Rabi, Summer, Winter and Whole Year observations. In the Indian context, Kharif refers to the monsoon cropping season, with crops typically sown from June–July and harvested around September–October, while Rabi refers to the winter season, with crops sown from October–November and harvested around April–May. Whole Year and Winter categories show the highest average yields, followed by Summer, while Kharif, Rabi and Autumn have lower mean yields. Figure A3 in Appendix A illustrates these seasonal differences, which support treating Season as an important categorical predictor in the modelling stage.

Figure 2 – Mean yield by state (spatial pattern)

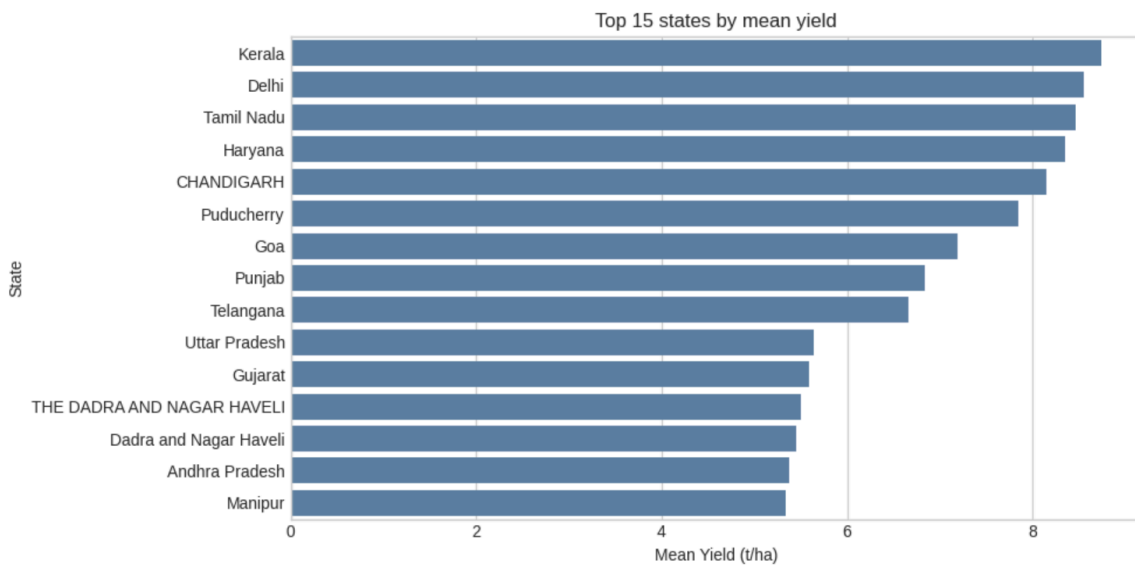


Figure 2: Mean yield (t/ha) by state in the cleaned APY dataset, highlighting substantial spatial heterogeneity in average productivity.

Mean yield by state in the cleaned APY dataset highlights substantial spatial heterogeneity in average productivity. Computing mean yields for each state and ranking them by their average productivity shows that several states and union territories such as Kerala, Delhi and Tamil Nadu have mean yields around or above 8 t/ha, while many others lie closer to or below the national average. Figure 2 illustrates these differences, with a small group of high-productivity states standing out clearly from the majority. These contrasts likely reflect a combination of agro-climatic conditions, cropping patterns and management practices, and they underline the importance of including State, and indirectly District, as key categorical predictors so that the models can learn region-specific yield levels.

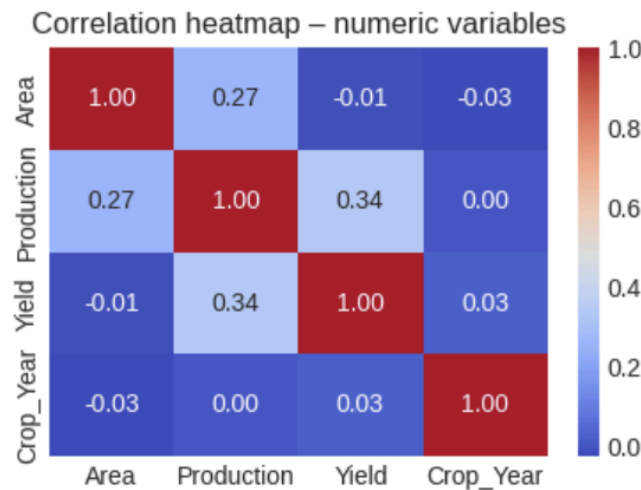


Figure 3 – Correlation heatmap

In addition to univariate summaries, as in figure 3 a correlation heatmap for the numerical variables (Area, Production, Yield and Crop_Year) shows that Production is moderately positively correlated with Yield (correlation ≈ 0.34), while Area has only a weak relationship with Yield and a modest positive correlation with Production. Crop_Year is essentially uncorrelated with the other numeric variables. This reinforces the view that Production largely reflects the algebraic link between area and yield and motivates excluding it from the predictive feature set to avoid target leakage, while still using the heatmap as a descriptive check on the structure of the cleaned data

Overall, the exploratory analysis indicates that yields vary strongly across seasons, crops, states and years, and that once yield has been derived, the raw area and production values add limited additional information for prediction beyond their role in defining yield. To explore variation between crops, mean yields are computed for each crop, focusing on the 15 with the largest number of observations. Figure A2 in Appendix A illustrates the wide range in typical yields, from low-yielding cereals and pulses to very high-yielding crops such as sugarcane and coconut, which occupy the upper tail of the distribution and later motivate the dedicated high-yield crop experiment. These insights guide the modelling design: the feature set is centred on the main categorical drivers (state, district, crop, season) alongside crop year and cultivated area, and a temporal evaluation strategy that respects the ordering of years is adopted.

4. Methodology

In this section yield prediction is formulated as a supervised regression task, a temporal train–validation–test split is constructed, the feature matrix is built, and the regression models and interpretability analyses used in the project are described. The emphasis is on the concrete design decisions made for the APY dataset rather than general machine-learning theory.

4.1 Problem formulation and evaluation metrics

Each cleaned APY record corresponds to a combination of state, district, crop, season and crop year, with numerical variables for area, production and yield. Yield (tonnes per hectare) is treated as a continuous target, and the goal is to learn a mapping from a feature vector describing location, crop, season, year and scale to this scalar yield value. This is therefore a supervised regression problem on tabular data.

Performance is assessed using three standard regression metrics: root mean squared error (RMSE), mean absolute error (MAE) and the coefficient of determination R^2 . RMSE and MAE are reported in tonnes per hectare and provide an absolute sense of prediction error, while R^2 measures the proportion of variance in observed yields explained by the model relative to a constant-mean baseline. All three metrics are computed on the training, validation and test sets to assess both overall accuracy and potential overfitting.

4.2 Temporal train–validation–test split

Because one intended use of the models is to forecast yields in future seasons, evaluation must respect the temporal structure of the data. A random row-wise split would allow information from later years to influence models used to predict earlier years and would likely overstate performance.

After cleaning, the dataset spans crop years 1997–2020. A temporal split is therefore defined: records from 1997–2012 form the training set, 2013–2015 the validation set and 2016–2019 the test set, with 2020 excluded from modelling because it shows an unexplained drop in average yield and may represent atypical conditions. This split results in 217,311 training rows, 47,036 validation rows and 71,331 test rows, and the notebook verifies that each record belongs to exactly one split. The validation set is used for model selection and hyperparameter choices, while the test set is reserved for the final performance comparison. Working with more than 300,000 observations and high-cardinality categorical variables also constrains the choice of models and encoders and requires careful attention to computational efficiency and memory use.

4.3 Feature set and encoding

The cleaned dataset `df_model` contains seven main variables: State, District, Crop, Season, Crop_Year, Area and Yield, together with Production, which is retained only for exploratory analysis. For modelling, State, District, Crop and Season are treated as categorical predictors, and Area and Crop_Year as numerical predictors, with Yield as the target. Production is **excluded** from the predictive feature set because it is mechanically related to Yield ($\text{production} = \text{area} \times \text{yield}$) and would introduce redundancy and target leakage into the models.

Most scikit-learn models do not accept string categories directly, so the four categorical variables are encoded using an OrdinalEncoder. The encoder is fitted on the training set only and the fitted mapping is then applied to transform the validation and test sets. To handle categories that appear only in later years, `handle_unknown="use_encoded_value"` and `unknown_value=-1` are used so that any unseen state, district, crop or season at prediction time is mapped to a special code rather than causing an error. For each split, the encoded categorical features and numerical features are horizontally stacked to construct the design matrices, which have shapes $217,311 \times 6$ (train), $47,036 \times 6$ (validation) and $71,331 \times 6$ (test). The corresponding target vectors contain the cleaned yield values.

4.4 Baseline linear model: Ridge regression

As a first benchmark a regularised linear model is trained using Ridge regression. Ridge extends ordinary least squares by adding an L^2 penalty on the regression coefficients, which helps stabilise the estimates when predictors are correlated. Here Ridge mainly serves to quantify how much of the variation in yields can be explained by a simple additive model in the encoded features.

Scikit-learn's Ridge is used with regularisation strength `alpha = 1.0` and `random_state = 42`. The model is fitted on the training design matrix and target vector, and RMSE, MAE and R^2 are then computed on the training, validation and test sets using a common evaluation helper in the notebook. The resulting metrics, reported in Section 5, show that the linear model captures only a modest fraction of the yield variability.

4.5 Tree-based ensemble models

The main focus of the project is on tree-based ensemble models, which are well suited to mixed categorical–numerical data and can capture non-linear interactions between features. Two such models are implemented: a Random Forest and a Histogram-based Gradient Boosting regressor.

For the Random Forest, scikit-learn's `RandomForestRegressor` is used with 200 trees (`n_estimators = 200`), unrestricted depth (`max_depth = None`), a minimum of five samples per leaf (`min_samples_leaf = 5`) and `n_jobs = -1` to use all available CPU cores. This configuration aims to balance flexibility against overfitting: a relatively large ensemble of unconstrained trees with a small minimum leaf size can capture fine-grained patterns in the APY data, while the ensemble averaging helps to stabilise predictions. The model is fitted on the training set and evaluated on the validation and test sets using RMSE, MAE and R^2 .

For the boosting approach, scikit-learn's `HistGradientBoostingRegressor` is adopted, which uses feature binning to speed up training on large tabular datasets. The baseline configuration uses `max_depth = 10`, `learning_rate = 0.1`, `max_iter = 200` and `min_samples_leaf = 20`. These values were chosen as a reasonable starting point based on scikit-learn guidance and the literature, with the aim of providing a fairly expressive but not excessively complex model. As with the Random Forest, the model is trained on the full training set and evaluated on the validation and test sets; validation performance and learning-curve diagnostics are used to check for overfitting.

4.6 Learning-curve diagnostics

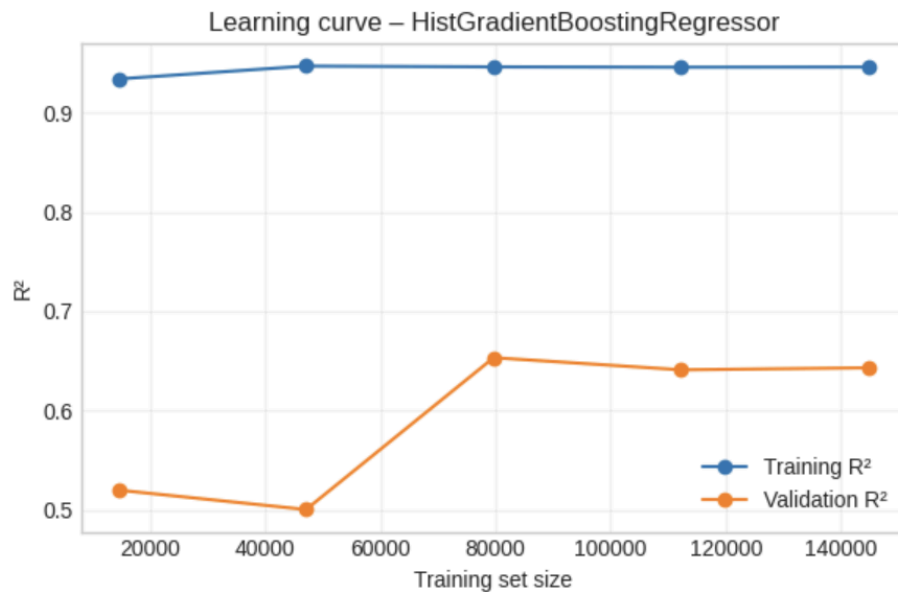


Figure 4: Learning curve for the Histogram-based Gradient Boosting model, with training and cross-validated R^2 plotted against training set size. Validation R^2 rises from around 0.5 to roughly 0.65 as more data are used, while training R^2 stays close to 0.95, indicating moderate overfitting but clear gains from additional data.

The learning curve in Figure 4 summarises how the Histogram-based Gradient Boosting model scales with data. Training R^2 is high and almost flat across all sample sizes, whereas validation R^2 improves substantially between the smallest and largest subsets and then begins to level off. The persistent gap between the two curves shows that the model does overfit to some extent, but the upward trend in validation performance suggests that adding more observations still improves generalisation rather than merely reinforcing noise.

4.7 Model refinement experiments

To demonstrate optimisation effort and explore the limits of the tree-based models, three refinement experiments are run.

The first tests a more strongly regularised Random Forest. Starting from the baseline configuration, the number of trees is increased to 300, `max_depth` is set to 20 and `min_samples_leaf` is raised to 10. These changes are intended to reduce overfitting by making each tree shallower and each leaf represent more observations, possibly at the cost of slightly higher bias. This model is trained on the same training data and its validation and test metrics are compared with those of the baseline Random Forest.

The second experiment targets crops with very high mean yields, such as sugarcane, coconut, banana, tapioca, potato and onion. The exploratory analysis showed that these crops occupy the upper tail of the yield distribution and may behave differently from typical cereals and pulses. To investigate whether a crop-specific model can better capture this regime, the train, validation and test sets are filtered to these crops and new feature matrices are built using the existing encoder and feature lists. A `RandomForestRegressor` with 200 trees and `min_samples_leaf` = 5 is then trained on the high-yield training subset and evaluated on the corresponding validation and test subsets. Finally, the global baseline Random Forest is applied to the same high-yield test data to see whether the specialised model offers any improvement.

The third refinement is a hyperparameter-tuning experiment for the gradient-boosting model. A `RandomizedSearchCV` procedure is run over `max_depth`, `learning_rate`, `max_iter` and `min_samples_leaf`, using 3-fold cross-validation and negative RMSE as the scoring metric. The best configuration found (shallower trees and lower learning rate) is then refitted on the full training set and its test performance is compared with that of the baseline `HistGradientBoosting` model. This experiment demonstrates the effect of more aggressive regularisation and tuning on predictive accuracy.

4.8 Feature importance and interpretability

To interpret the fitted models and understand which variables drive the predictions, feature-importance analyses are carried out for both the Random Forest and the Histogram-based Gradient Boosting model.

Figure 5 – Feature importance (Random Forest)

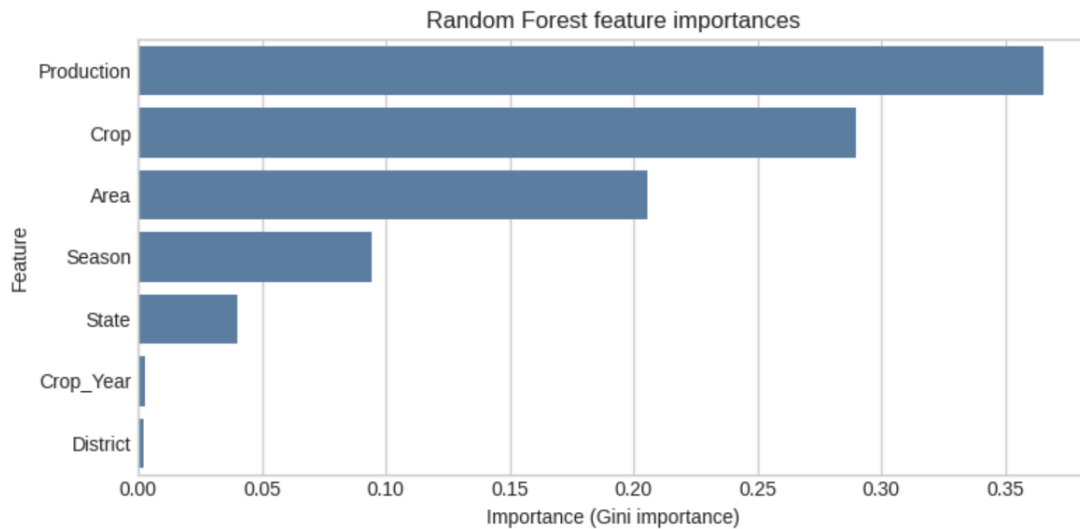


Figure 5: Impurity-based feature importances for the Random Forest model, showing Crop as by far the most influential predictor, followed by State and Season, with Area contributing a smaller effect and Crop_Year and District playing only minor roles

For the Random Forest, the built-in impurity-based importance scores are used; these measure the total reduction in the splitting criterion attributable to each feature across all trees in the ensemble. Scores are computed for all six predictors and visualised as a horizontal bar plot. The resulting importances shows Crop as the most influential feature, followed by State and Season, with Area playing a smaller but non-negligible role and Crop_Year and District contributing least.

For the gradient-boosting model, which does not expose impurity-based importances in the same way, permutation importance on the test set is used. In this approach each feature is randomly shuffled in turn while holding others fixed, and the resulting drop in test-set R^2 is used as a measure of importance. The permutation-based importances for the Histogram-based Gradient Boosting model again highlight Crop as the dominant predictor, with Season and State causing noticeable reductions in R^2 when shuffled, Area having a much smaller effect, and District and Crop_Year having almost no impact on performance. These results quantify how much each feature contributes to predictive performance in the final model. Together with residual plots by crop and state, they are used in the Discussion to relate model behaviour back to agronomic intuition and to identify where the models still perform poorly.

5. Results

This section summarises the performance of the models on the training, validation and test sets using RMSE, MAE and R^2 . The three main models (Ridge, Random Forest and Histogram-based Gradient Boosting) are compared first, followed by the regularised Random Forest and the high-yield crop experiment, and finally an overall preferred model is selected.

5.1 Main models

The Ridge regression model acts as a linear baseline. On the training set it achieves an RMSE of approximately 11.20 t/ha, an MAE of 5.32 t/ha and an R^2 of 0.11; on the test set RMSE increases to about 12.20 t/ha, MAE to 5.68 t/ha and R^2 falls to 0.09. These numbers confirm that a linear model with additive effects of the encoded predictors captures only a small fraction of the variation in district-level yields, and the similar training and test errors indicate high bias rather than overfitting.

The Random Forest regressor performs substantially better. With 200 trees and a minimum leaf size of five, it attains a training RMSE of about 2.11 t/ha and an R^2 of 0.97, indicating a very close fit to the training data. On the validation set the RMSE increases to roughly 4.23 t/ha and R^2 decreases to 0.89, while on the test set the RMSE is around 5.78 t/ha, the MAE is 1.77 t/ha and R^2 remains reasonably high at 0.79. The gap between training and test errors shows some overfitting, but generalisation is still strong enough to demonstrate that non-linear ensembles of trees can model interactions in the APY data much more effectively than Ridge regression.

The Histogram-based Gradient Boosting model delivers the best performance among the main models. On the training set it achieves an RMSE of about 2.93 t/ha, an MAE of 1.06 t/ha and an R^2 of 0.94, slightly less extreme than the Random Forest but still indicating a close fit. On the test set it achieves a lower RMSE of approximately 5.41 t/ha, an MAE of 1.87 t/ha and a higher R^2 of 0.82. Relative to the Random Forest, the boosted model reduces test RMSE by about 0.37 t/ha and increases R^2 by roughly three percentage points, consistent with the expectation that gradient-boosted trees can extract slightly more predictive signal from tabular data when appropriately configured.

Figure 6 visualises the relationship between predicted and observed yields for this best-performing Histogram-based Gradient Boosting model on the test set. The tight band of points around the 1:1 line at moderate yields is consistent with the RMSE of 5.41 t/ha and R^2 of 0.82, whereas the widening scatter for very high yields confirms that the model struggles most in the upper tail of the distribution, particularly for high-yield crops such as sugarcane and coconut. Each point represents one test observation, with observed yield on the x-axis and model prediction on the y-axis.

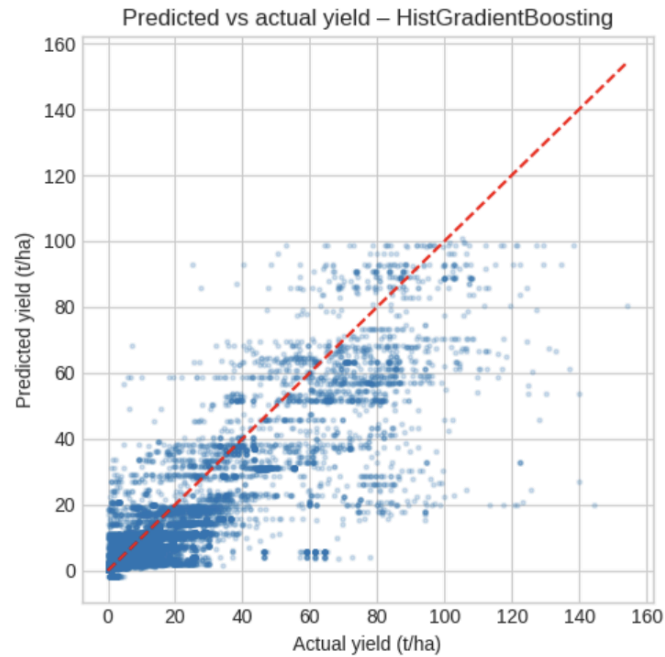


Figure 6: Predicted versus actual district-level yields (t/ha) on the held-out test set for the Histogram-based Gradient Boosting model, with the dashed line showing the 1:1 reference. Most points cluster around the diagonal, especially in the 2–8 t/ha range, indicating generally accurate predictions, while the increasing spread at very high yields highlights larger absolute errors for extreme observations.

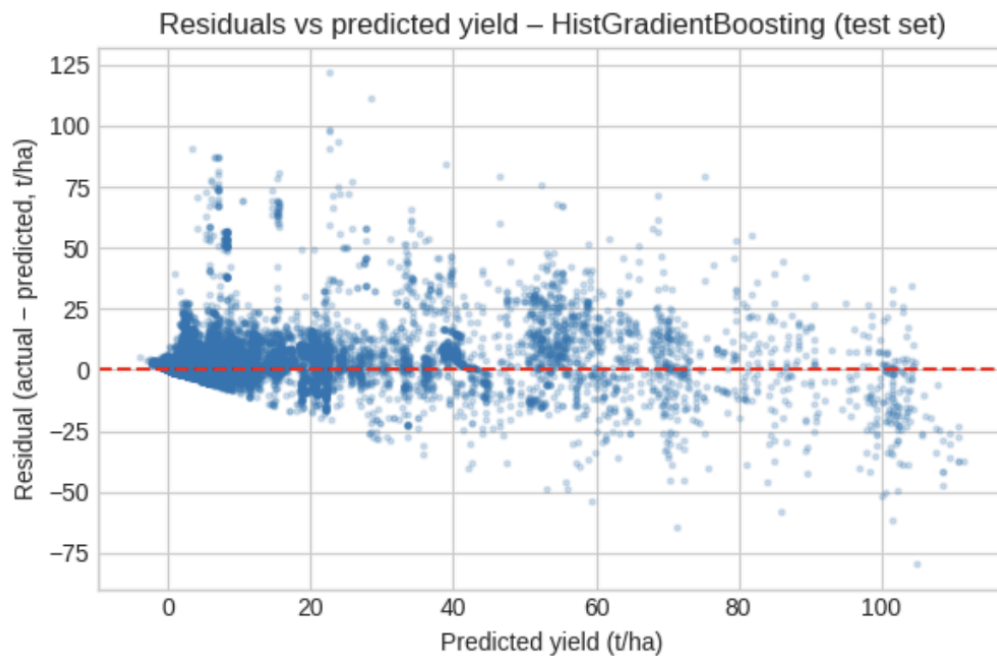


Figure 7: Residuals (actual – predicted yield, t/ha) versus predicted yield on the held-out test set for the Histogram-based Gradient Boosting model, with the dashed line indicating zero error.

The residual plot in Figure 7 further examines how the Gradient Boosting model behaves across the yield range. For low and moderate predicted yields, most residuals lie close to zero, suggesting that the model is approximately unbiased for the bulk of districts and that typical errors remain small, in line with the overall test RMSE. As predicted yields increase, the vertical spread of residuals widens and several large positive residuals appear, indicating that the model tends to under-predict some very high-yield observations and is less accurate in the extreme upper tail of the distribution.

5.2 Regularised Random Forest

The regularised Random Forest tests whether stronger constraints on tree depth and leaf size can reduce overfitting. With 300 trees, `max_depth = 20` and `min_samples_leaf = 10`, it obtains a training RMSE of about 2.57 t/ha, an MAE of 0.69 t/ha and an R^2 of 0.95, indicating that it still fits the training data closely but slightly less aggressively than the baseline Random Forest. On the test set, RMSE rises to approximately 5.85 t/ha, MAE to 1.80 t/ha and R^2 is about 0.79, all very similar to but marginally worse than the baseline Random Forest and clearly inferior to the Histogram-based Gradient Boosting model. This suggests that the original Random Forest configuration already provided a reasonable balance between variance and bias, and that simply tightening regularisation in this way does not improve generalisation.

5.3 High-yield crops experiment

The high-yield crop experiment focuses on sugarcane, coconut, banana, tapioca, potato and onion, which the exploratory analysis identified as having much higher average yields than most cereals and pulses. Filtering the train, validation and test sets to these crops yields subsets of 24,521, 5,524 and 8,302 observations respectively. On this restricted data a Random Forest with the same configuration as the baseline model is trained.

On the high-yield training subset the crop-specific Random Forest achieves low in-sample error (RMSE $\approx$ 5.94 t/ha, MAE $\approx$ 3.00 t/ha, $R^2 \approx$ 0.95), but performance degrades on the high-yield test subset, where RMSE increases to about 13.39 t/ha, MAE to 8.03 t/ha and R^2 drops to 0.74. Applying the global baseline Random Forest, trained on all crops, to the same high-yield test data yields a higher RMSE of roughly 15.75 t/ha, a higher MAE of 8.86 t/ha and a lower R^2 of 0.64. In other words, the crop-specific Random Forest does outperform the global model on the high-yield subset, but both models still exhibit large errors for these extreme yields, indicating that accurately predicting the upper tail remains challenging.

5.4 Overall comparison

Table 1 below summarises the test-set performance of the four main models.

Model	Test RMSE (t/ha)	Test MAE (t/ha)	Test R ²
Ridge (linear baseline)	12.20	5.68	0.09
Random Forest	5.78	1.77	0.79
Regularised Random Forest	5.85	1.80	0.79
HistGradientBoosting (HGB)	5.41	1.87	0.82

As Table 1 shows, the Ridge baseline performs worst by a large margin, while the Random Forest and its regularised variant substantially reduce error relative to the linear model. The Histogram-based Gradient Boosting model achieves the lowest RMSE and highest R² on unseen years, making it the strongest of the configurations tested.

To summarise the test results, the Ridge model performs worst by a clear margin, with RMSE of approximately 12.20 t/ha, MAE of 5.68 t/ha and R² of 0.09. The baseline Random Forest dramatically improves on this, with RMSE of about 5.78 t/ha, MAE of 1.77 t/ha and R² of 0.79, while the regularised Random Forest is slightly weaker with RMSE around 5.85 t/ha and a similar R². The Histogram-based Gradient Boosting model achieves the best overall performance, with RMSE of roughly 5.41 t/ha, MAE of 1.87 t/ha and R² of 0.82 on unseen years. Based on this comparison I select the Histogram-based Gradient Boosting model as the preferred model for the project, using the Random Forest primarily for cross-checking feature-importance patterns and for the high-yield crop experiment.

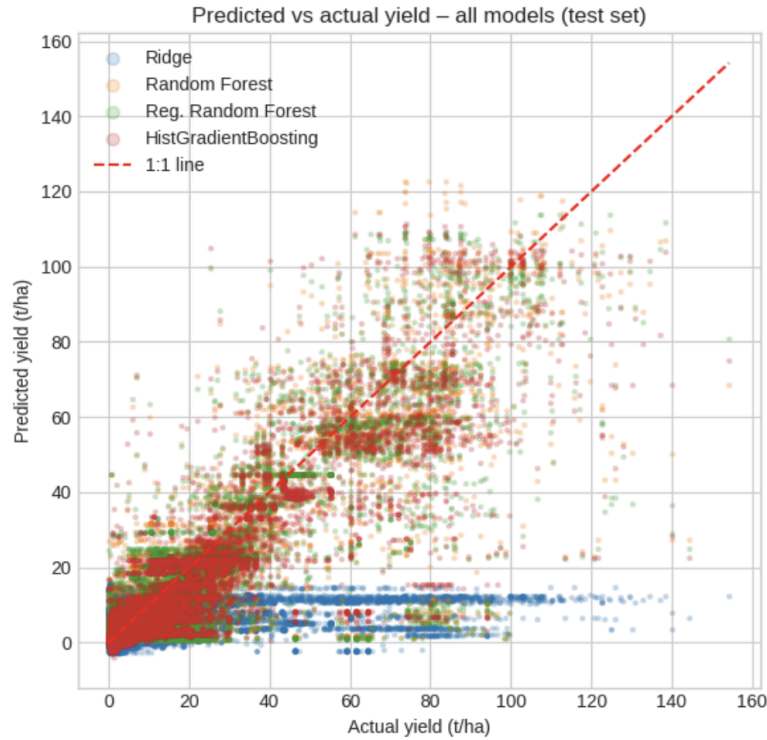


Figure 8: Predicted versus actual district-level yields (t/ha) on the held-out test set for all four models, with the dashed line showing the 1:1 reference. Ridge predictions form a narrow, almost flat band well below the diagonal, whereas the three tree-based models cluster much closer to the 1:1 line, with the Histogram-based Gradient Boosting points lying closest overall.

Figure 8 provides a visual comparison of test-set predictions across all four models. The Ridge regression points remain concentrated between roughly 0 and 15 t/ha even when observed yields are much higher, confirming that the linear baseline under-fits the data and explains only a small fraction of the variance, in line with its low test R^2 of about 0.09. By contrast, the Random Forest, regularised Random Forest and Histogram-based Gradient Boosting predictions follow the 1:1 line far more closely, mirroring their substantially lower RMSE and higher R^2 values. Among these tree-based methods, the boosted model produces the tightest band of points around the diagonal, especially for typical yields, supporting its selection as the preferred model for district-level yield prediction on the APY dataset.

6. Discussion and conclusion

Overall, the results show that tree-based machine-learning models trained solely on APY statistics can predict district-level crop yields in India with moderate to high accuracy (test R^2 up to about 0.82 and RMSE around 5.4 t/ha), and that Histogram-based Gradient Boosting performs clearly better than the linear Ridge baseline and slightly better than the Random Forest. This directly answers the research question about how far tree-based models can go when restricted to administrative APY data and how they compare with a linear alternative. The Ridge regression model explains only a small share of the variance in yields and has similar errors on training and test sets, indicating that an additive relationship in the encoded predictors is too restrictive for the complex interactions between crop, location, season, area and crop year. In contrast, both the Random Forest and the Histogram-based Gradient Boosting model achieve much higher R^2 values and substantially lower RMSE on unseen years, and their predicted-vs-actual plots in Figures 6 and 8 show points that track the 1:1 line much more closely, demonstrating that tree ensembles can exploit strong structure in administrative agricultural statistics even without explicit weather or satellite inputs.

Feature-importance analyses help explain what the best models have learned. For the Random Forest, impurity-based importances (Figure 5) show that Crop is by far the most influential feature, followed by State and Season, with Area playing a smaller but non-negligible role and Crop_Year and District contributing relatively little. Permutation importance for the Histogram-based Gradient Boosting model (Figure A4) leads to a similar ranking: shuffling Crop causes the largest drop in test-set R^2 , followed by shuffling Season and State, whereas shuffling Area has a smaller effect and shuffling Crop_Year or District has almost none. This confirms that both models rely primarily on crop identity and broad seasonal and spatial context, with cultivated area providing some additional information, while simple time trends and fine-grained district identifiers add relatively little marginal predictive power.

Between the two main ensemble models, Histogram-based Gradient Boosting consistently outperforms Random Forest by a modest but meaningful margin in terms of test RMSE and R^2 , and Figures 6–8 show that its predictions lie closest to the 1:1 line across the yield range. This pattern is consistent with the findings of Singh and Gupta (2022), who also report that gradient-boosting models achieve slightly better test performance than Random Forests for district-level yields in India when appropriately tuned. Both models fit the training data well, but the boosted model generalises slightly better to later years, suggesting that its sequential error-correcting structure is more effective at capturing subtle interactions in the APY features. The regularised Random Forest does not improve on the baseline configuration and performs marginally worse on the test set, indicating that simply constraining tree depth and leaf size did not provide a better bias–variance trade-off for this problem. The high-yield crop experiment shows that training a separate Random Forest on a small subset of high-yield crops does outperform the global model on that subset but still leaves large absolute errors, indicating that accurately predicting the very highest yields remains challenging even with specialised models. A targeted hyperparameter search for the Histogram-based Gradient Boosting model using randomised search confirms that more aggressive regularisation (shallower trees and lower learning rate) can actually worsen test performance relative to the chosen baseline, so the final configuration reflects a compromise between complexity and generalisation rather than exhaustive optimisation.

Several limitations qualify how these findings should be interpreted. First, the APY dataset is an administrative product with measurement error, extreme outliers and aggregation over heterogeneous farms within each district; even after careful cleaning using crop-specific yield caps, the models inevitably learn patterns in reported statistics rather than in underlying biophysical processes. Second,

the feature set is intentionally restricted to routinely collected variables and does not include gridded weather or remote-sensing indices, so the models cannot directly respond to climatic anomalies or management changes beyond those already embedded in past APY values. Third, most of the modelling uses a single national model, and residual patterns for specific crops and states indicate that some crop–region combinations remain systematically over- or under-predicted. Finally, although one regularised Random Forest, a high-yield crop model and a randomised hyperparameter search for the gradient-boosting model are explored, the study does not carry out an exhaustive search over model families or tuning spaces.

An inspection of residuals by crop and state in the notebook shows that errors tend to be larger and sometimes biased for the small group of very high-yield crops and for a few states with extreme mean yields; the residual-versus-predicted plot in Figure 7 similarly shows widening residual spread at high predicted yields. This is consistent with the high-yield crop experiment, where both the crop-specific and global Random Forests exhibit sizeable test errors despite reasonable R^2 values, and it supports the idea that future work should explore hierarchical or crop-specific models for these more difficult regimes. Given the scale and heterogeneity of the APY data and the constraint of using APY-only features, achieving test R^2 values around 0.8 suggests that the chosen modelling strategy captures much of the predictable structure in the dataset, while still leaving room for improvement through richer covariates and more tailored model architectures.

These limitations motivate several directions for future work. A natural extension would be to integrate additional covariates, such as seasonal rainfall, temperature and satellite-derived vegetation indices, and to quantify how much they reduce error beyond what can be achieved with APY alone. Another line of work would develop crop- or region-specific models within a hierarchical or multi-task framework, allowing the model to share information across crops while still adapting to local behaviour. It would also be valuable to move from point predictions to probabilistic forecasts, using approaches such as quantile regression or Bayesian ensembles so that decision makers can explicitly consider uncertainty in predicted yields. Combining the Histogram-based Gradient Boosting model with conformal prediction or other calibration techniques to generate empirically valid prediction intervals for district-level yields—for example, “90% chance yield lies between 3 and 6 t/ha”—would allow planners to work with risk bands rather than single-value forecasts. Taken together, these extensions would help turn the modelling framework developed here into a more robust and operationally useful yield-prediction system for India and other data-constrained settings.

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Appendix

Appendix A: Additional figures

Figure A1: Distributions of the numerical variables Area, Production and Yield in the raw APY dataset (after cleaning), showing strong right-skew and extreme upper tails for Production and Yield.

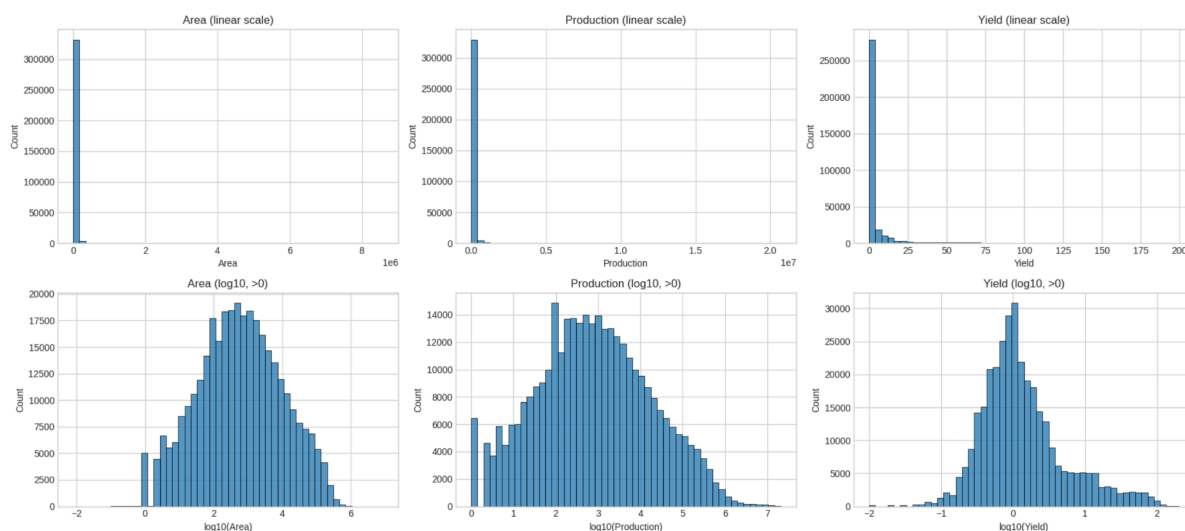


Figure A2: Mean yield (t/ha) by crop for the 15 crops with the largest number of observations, illustrating the wide range in typical productivity between cereals, pulses and high-yield crops such as sugarcane and coconut.

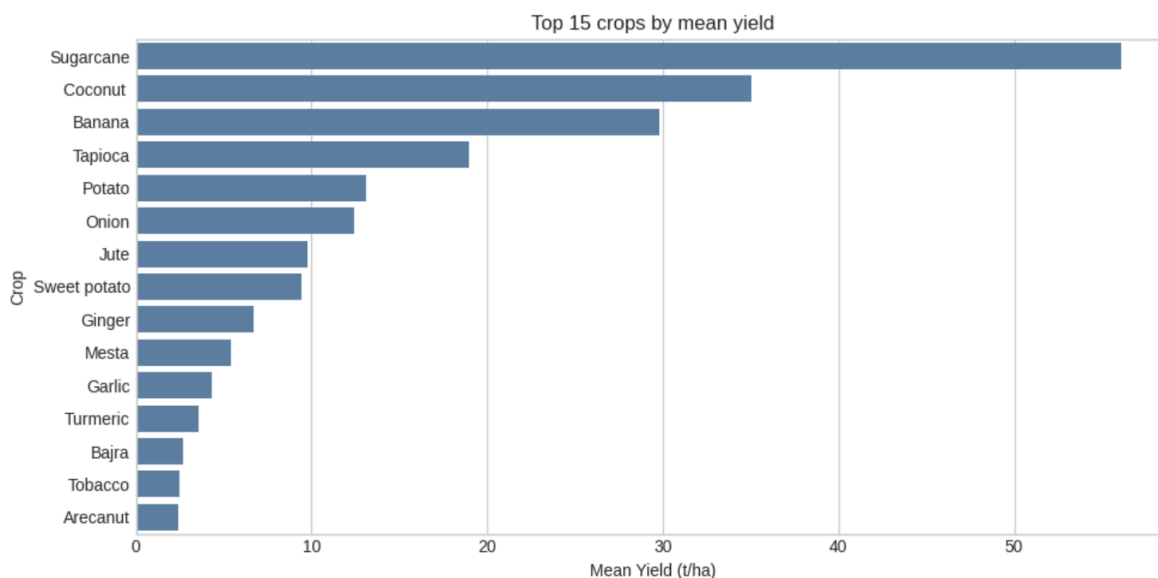


Figure A3: Mean yield (t/ha) by season, highlighting clear differences between Kharif, Rabi and other seasons in the APY data.

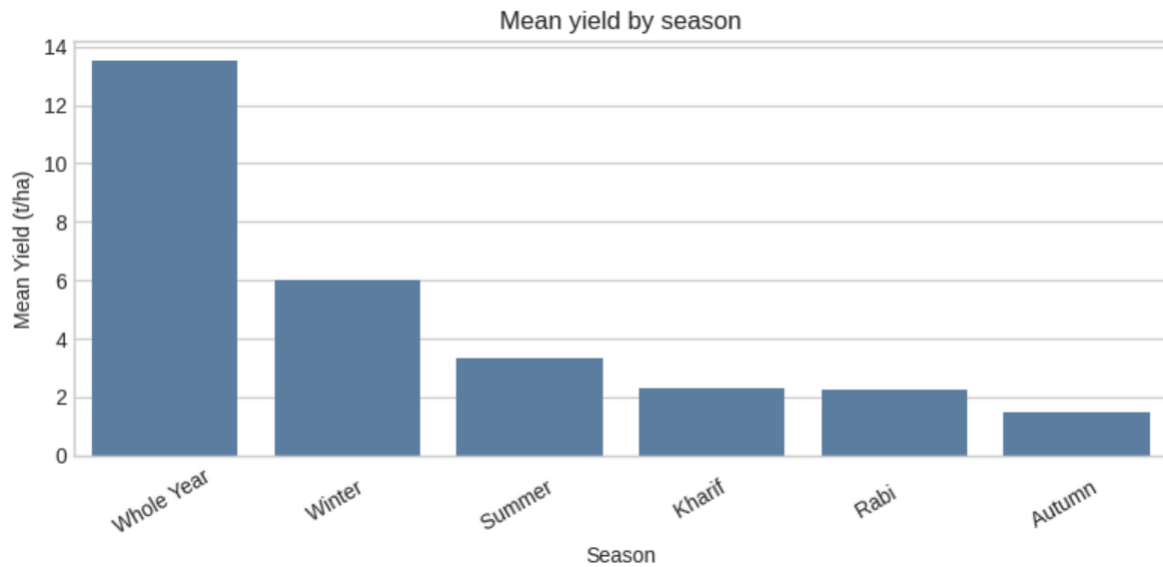
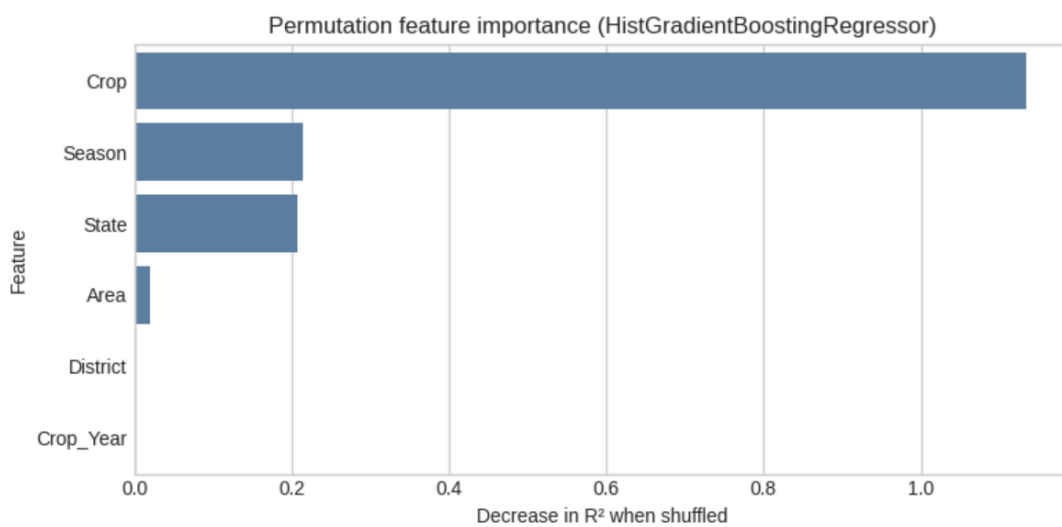


Figure A4: Permutation-based feature importances for the Histogram-based Gradient Boosting model, measured as the drop in test-set R^2 when each feature is shuffled, showing Crop as the dominant predictor, followed by Season and State, with Area having a smaller effect and Crop_Year and District contributing very little.



Appendix B: Code and GitHub repository

This appendix summarises the main code artefacts used in the project and provides the link to the full GitHub repository containing all versions of the code.

B.1 GitHub repository

All code, notebooks and configuration files for this project are available in the following public GitHub repository:

- **GitHub URL:** https://github.com/vamshi-34/Crop_Prediction